

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

2 4

Intensity Frequency

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

[10 10 10, 10 20 10, 10 10 10]

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$\begin{bmatrix} -1 & -2 & -1, & 0 & 0 & 0, & 1 & 2 & 1 \end{bmatrix}$

to:

[10 10 10, 20 20 20, 30 30 30]

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0	4
1	6
2	10
3	6

- (a) Interpret the need for equalization.
- (b) Justify use of CDF.
- (c) Compute cumulative distribution.
- (d) Compute new intensity levels.
- (e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

- (a) Interpret noise characteristics.
- (b) Justify median filter usage.
- (c) Apply filter to:

10 200 10,200 50 200,10 200 10

- (d) Compute center value.
- (e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

- (a) Interpret smoothing objective.
- (b) Justify Gaussian filter over averaging.
- (c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10,20 40 20,10 20 10]

- (d) Compute center pixel.
- (e) Interpret smoothing effect.

16. High-Pass Filtering

An image requires sharpening.

- (a) Interpret need for high-pass filtering.
- (b) Justify filter selection.

(c) Apply mask:

$[-1 \quad -1 \quad -1, -1 \quad 8 \quad -1, -1 \quad -1 \quad -1]$

(d) Compute center value.

(e) Interpret edge enhancement.

17. Frequency Domain High-Pass

Given frequency components:

$$F = [10, 20, 80, 100]$$

(a) Interpret high-frequency importance.

(b) Justify high-pass filtering.

(c) Apply filtering (remove low frequencies).

(d) Compute output.

(e) Interpret result.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$[-1 \quad 0 \quad 1, -2 \quad 0 \quad 2, -1 \quad 0 \quad 1]$$

Matrix:

$$[10 \quad 20 \quad 30, 10 \quad 20 \quad 30, 10 \quad 20 \quad 30]$$

(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

20. Prewitt Edge Detection

(a) Interpret Prewitt operator purpose.

(b) Justify use over Sobel.

(c) Apply kernel:

$[-1 \ 0 \ 1, -1 \ 0 \ 1, -1 \ 0 \ 1]$

(d) Compute output.

(e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

(a) Interpret segmentation problem.

(b) Justify splitting method.

(c) Apply threshold 80.

(d) Compute regions.

(e) Interpret separation.

22. Region Merging

Regions: [45, 50], [52, 55]

(a) Interpret merging criteria.

(b) Justify merging approach.

(c) Apply threshold = 5.

(d) Compute merged region.

(e) Interpret result.

23. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

(a) Interpret problem of discontinuity.

(b) Justify linking method.

(c) Apply linking step.

(d) Compute result.

(e) Interpret continuity.

24. Low-Pass Filtering (Frequency)

Given $F = [200, 150, 50, 20]$

(a) Interpret smoothing requirement.

(b) Justify LPF selection.

(c) Apply filter.

- (d) Compute output.
- (e) Interpret effect.

25. Laplacian Sharpening (Variant)

Mask:

$$\begin{array}{ccc} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{array}$$

- (a) Interpret sharpening need.
- (b) Justify mask.
- (c) Apply to matrix.
- (d) Compute center.
- (e) Interpret result.

26. Histogram Matching

- (a) Interpret need for matching.
- (b) Justify method.
- (c) Map values [10, 50, 100].
- (d) Compute new values.
- (e) Interpret.

27. Noise Reduction Comparison

- (a) Interpret noise type.
- (b) Compare averaging vs median.
- (c) Apply both.
- (d) Compute outputs.
- (e) Interpret differences.

28. Combined Filtering

- (a) Interpret need for smoothing + sharpening.
- (b) Justify sequence.
- (c) Apply filters stepwise.
- (d) Compute output.
- (e) Interpret improvement.

29. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

- (a) Interpret multi-level segmentation.
- (b) Justify thresholds (50,100).
- (c) Apply segmentation.
- (d) Compute classes.
- (e) Interpret.

30. Boundary Detection

- (a) Interpret boundary extraction.
- (b) Justify method.
- (c) Apply edge + linking.
- (d) Compute result.
- (e) Interpret boundaries.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical problem solving
CO2- AT-1

SIVA (A) KODURU
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1. Grey level Transformation

(a) Improves image contrast by changing pixel intensity

(b) process:-

$$\Rightarrow s = 255 - r$$

$$\Rightarrow 0 \rightarrow 255$$

$$\Rightarrow 50 \rightarrow 205$$

$$\Rightarrow 100 \rightarrow 155$$

$$\Rightarrow 150 \rightarrow 105$$

$$\Rightarrow 255 \rightarrow 0$$

Ans: 255, 205, 155, 105, 0

2. log Transformation

(a) Brightens dark pixels

(b) process:-

$$* \ln(1) = 0$$

$$* \ln(2) = 0.693$$

$$* \ln(4) = 1.386$$

$$* \ln(8) = 2.079$$

Ans: 0, 0.693, 1.386, 2.079

3. Histogram Equation

(a) Improves image contrast

(b) process:-

$$* \text{total pixels} = 16$$

$$* \text{cdf} = 2, 8, 8, 16$$

92/100

Ans: 1, 2, 4, 7

4. Smoothing filter

① Removes filter noise

② Process:

* Add all 9 pixels

* Divide by 9

* Avg = 47

Ans: Center pixel = 47

③ Sharpening

④ Enhances edges

⑤ Process:

* Apply Laplacian mask

* Multiply mask with neighborhood

* Add values

Ans: Center pixel = 80

6. Frequency domain filter

① LPF smooths image; HPF sharpens image

② Process:

* Keep only first two frequency components

* Remove remaining frequency

Ans: Smooth image obtained.

Thresholding

② Separate foreground and Background

③ process Threshold = 100

$$* 60 < 100 \rightarrow 0$$

$$* 90 < 100 \rightarrow 0$$

$$* 120 \geq 100 \rightarrow 1$$

Ans: 0 0 1

8. Sobel-Edge Detection

② Detect edges

③ process: Apply Sobel kernel

* Multiply corresponding values

* Add results

Ans: gradient = 60

9. Region Growing

② Groups similar pixels

③ process: seed = 50

$$* |48 - 50| = 2 \checkmark$$

$$* |52 - 50| = 2 \checkmark$$

$$* |60 - 50| = 10 \times$$

Ans: Region = 48, 50, 52, 54

10. Smoothing + thresholding

(a) Remove noise before segmentation

(b) process: ≥ 1 brightness image

$$(c,d) = \sqrt{0} = 0 \quad T_{25} = 5$$

$$\sqrt{100} = 10$$

$$T_{225} = 15$$

Ans: 0, 5, 10, 15

11. Gamma Transformation

(a) Control Brightness

(b) process: mean = 48

Threshold = 50

$48 < 50 \rightarrow 0$

Ans $\rightarrow 0, 11$

12. Contrast stretching

(a) improve contrast

(b, c, d) + process:-

$$\# 50 \rightarrow 0$$

$$\# 75 \rightarrow 64$$

$$\# 100 \rightarrow 128$$

$$\# 150 \rightarrow 255$$

Ans: 0, 64, 128, 255

13. Histogram Equalization

- Ⓐ Improves image contrast
- Ⓑ CDF preserve intensity order
- Ⓒ * CDF $\Rightarrow 4, 10, 20, 26$
- Ⓓ * New level $= (\text{CDF}(26) \times 15)$
* New level $= 2, 6, 10, 15$
- Ⓔ Contrast is improved

14. Median filtering

- Ⓐ Removes salt and pepper noise
- Ⓑ preserve edges better than averaging
- Ⓒ process: 20, 24, 25, 26, 27, 28, 29, 30, 255
- Ⓓ Median $= 27$
- Ⓔ Noise is removed efficiently

15. Gaussian Smoothing

- Ⓐ Reduces image noise
- Ⓑ Giving more weight to nearby pixels
- Ⓒ process :- * Apply Gaussian kernel
* divide result by 16.

16. High pass filtering

- ① Enhance edges
- ② Highlight fine details
- ③ Apply high pass mask
- ④ center value = 80
- ⑤ Sharper image

17. Frequency domain high-pass

- ① High frequency contain edge information
- ② Remove smooth background
- ③ Remove low frequencies
- ④ output high frequency component only
- ⑤ Sharper results

18. Adaptive thresholding

- ① Handles uneven lighting
- ② use local threshold
- ③ threshold = 100
- ④ output: 0, 10, 11
- ⑤ Bkta segmentation

Vertical Sobel.

- ⑦ Detects vertical edges
- ⑧ Better noise suppression
- ⑨ Apply Sobel kernel
- ⑩ Centre value = 80
- ⑪ Strong vertical edges

⑧ Prewitt edge detection

- ① Prewitt edge detection
- ② Simple than Sobel
- ③ Apply prewitt kernel
- ④ Output = 80
- ⑤ Indicates vertical edge.

⑨ Region Merging

- ① Merge similar neighbouring regions
- ② Based on intensity difference
- ③ Difference = $G > 5$
- ④ No merging
- ⑤ Region remains separate

⑩ Edge Linking

- ① Connect broken edges

(1) Improves boundary continuity

(2) process :- original :- 10101

After linking :- 11111

(3) Result = 11111

(4) Continuous edges obtained

(5) Low-pass Filtering

(a) Remove high-frequency noise

(b) preserve low frequencies

(c) process :- $\times$ - remove high freq

(d) output = low frequency component

(6) Laplacian sharpening

(a) Enhances details

(b) Selects edges in all direction process

(c) Apply Laplacian mask

(d) center pixel = 80

(e) Sharper image.

Histogram matching

- (a) matches reference histogram
- (b) gives consistent appearance
- (c) process :-
 - $10 \rightarrow 20$
 - $50 \rightarrow 80$
 - $100 \rightarrow 160$
- (d) output :- 20, 80, 160
- (e) better Brightness

27. Noise Reduction Comparison

- (a) salt-and pepper noise

- (b) Avg :- sum/3
- Median Middle vs

- (c) Avg = 219

- (d) Median = 24

- (e) Median filter is better

28. Combined filtering

- (a) Removes noise than sharper

- (b) prevents noise amplification

- (c) process

Avg :- sharper

② Find output = 50

③ Better image quality

② Multi thread Segmentation

① divides image into multiple region

② user threshold 50 to 100

③ process: 20 < 50 $\rightarrow$ class 1

60 $\rightarrow$ class 2

100 $\rightarrow$ class 3

④ class: 1, 2, 3

⑤ better object separation

③ Boundary detection

① detect object boundary

② use edge and linking

③ process:

detected edge $\rightarrow$ link Broken edge

④ output: continuous boundary

⑤ improve object recognition